Stonemaier Games
Data Visualization Project

Dan Waxman
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Project Context
This data visualization project utilized data from 1000 latest plays in Board Game Arena in various games under Stonemaier Games. The goal of this project was to discover any new findings from the data that can help SM Games through data analysis and data visualization. The project also provides Dan Waxman with work products for other interested parties for future efforts in data analysis and data visualization.

Project Elements
The project includes three parts for each game: Description of the visualization, visualization, and key findings. The purpose of the description is to tell SM Games and other potential audiences about the visualization being used. The visualization depicts the information. Lastly, the key findings illustrates key findings found with the data.

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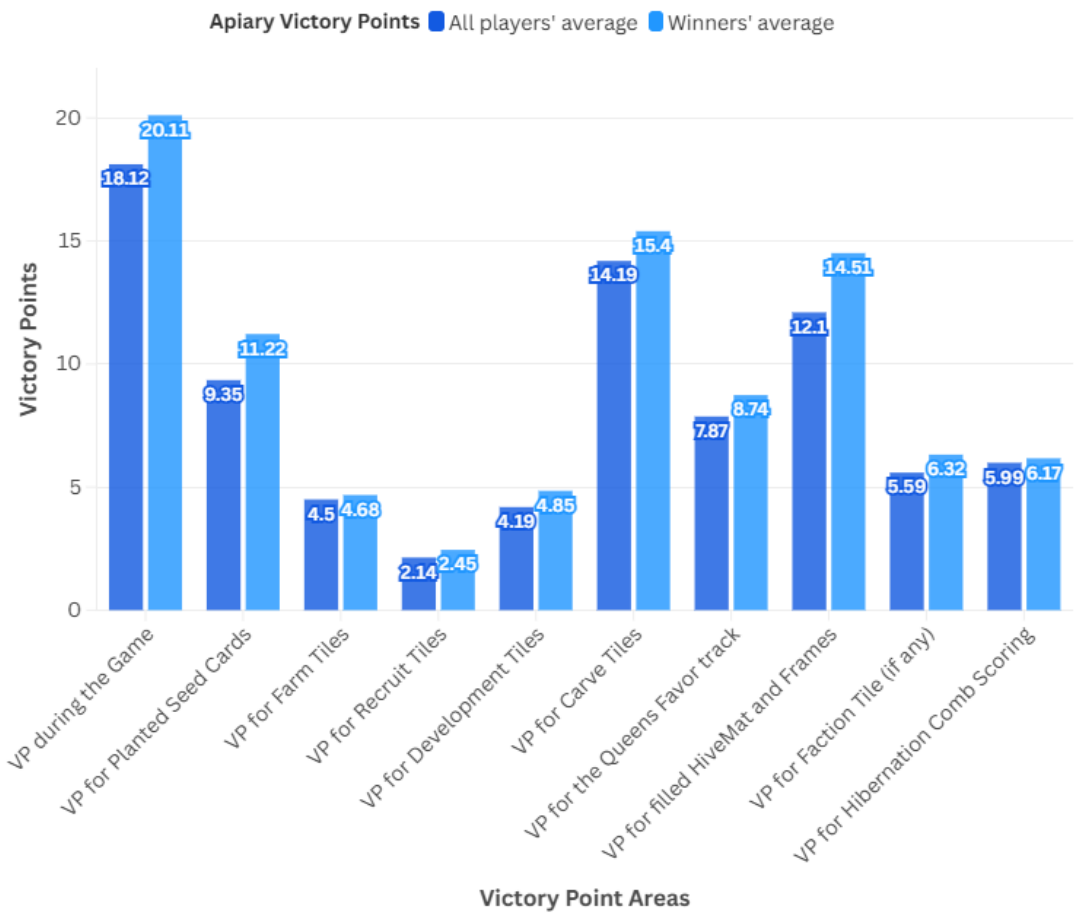
About SM Games
Stonemaier Games is a global leader in the Board Game Industry. (SM Games) strive(s) to bring joy to tabletops worldwide through games that are memorable, beautiful, and fun. SM Games was founded by Jamey Stegmaier and Alan Stone. SM games has a commitment to accessibility, diversity, and environmental sustainability. SM games have won many awards including for Connoisseur-gamer Game of the Year for Wingspan.

About Dan Waxman
Dan Waxman holds a BA from The University of North Carolina at Chapel Hill and a Master of Public Administration from George Mason University. He is currently pursuing data analytics and data visualization through studies with a certificate in Data Analytics, completing projects relating to data visualization, and coursework with DataCamp. He enjoys playing many board games both in-person and on Board Game Arena.

- Double bar chart created using *Flourish* to illustrate comparison.



Key Findings

- Winners tended to gain **more** victory points (VP) in three areas quite a bit more when compared with other areas: Planted Seed Cards, Carve Tiles, and HiveMat and Frames.
- VP for Hibernation Comb Scoring yielded the least difference between average players' average and Winners' average.

- Pictogram used to visually show the items that scored for an easy to see differences appeal. Used *Noun Project*, *Peoplegraphmaker*, and *Excel*.

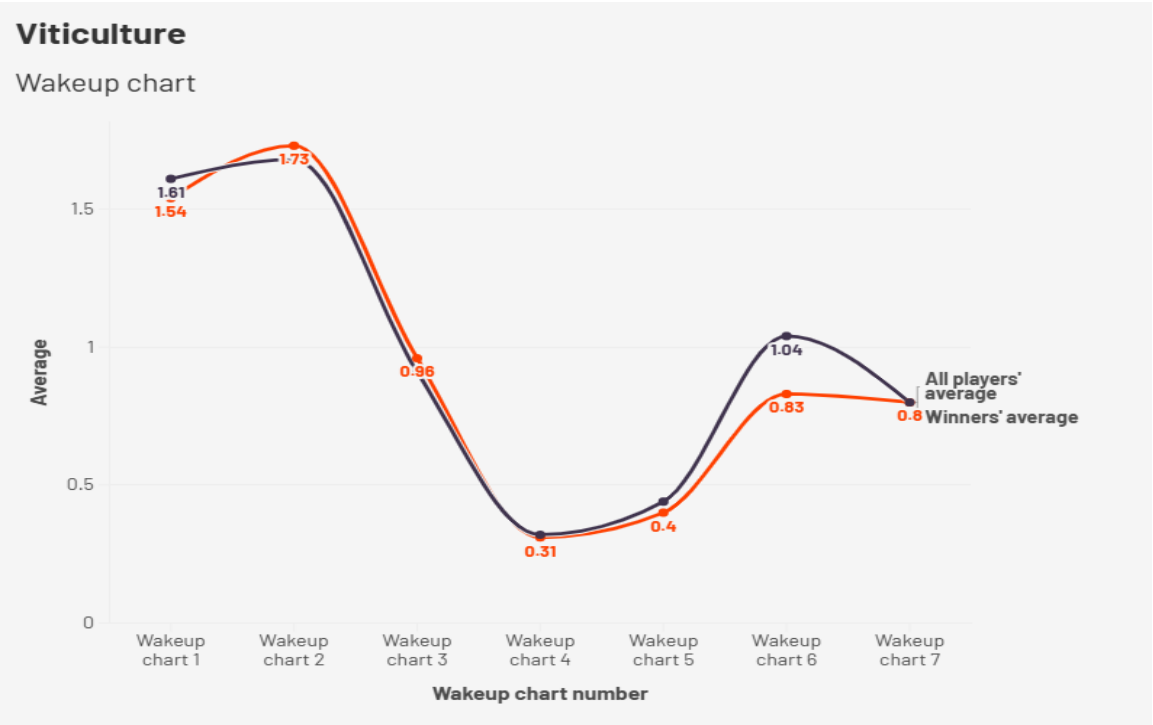
Scores from Different Game Elements

Score from stars	Average Players Average		14.96
	Winners' Average		22.85
Score from territories	Average Players Average		18.72
	Winners' Average		27.53
Score from resources	Average Players Average		3.52
	Winners' Average		4.53
Score from structure bonus tile	Average Players Average		2.61
	Winners' Average		3.03
Score from coins	Average Players Average		15.62
	Winners' Average		18.97

Key Findings

- Coins, stars, and territories yielded the most scoring.
- Structure bonus tile yielded the least scoring.

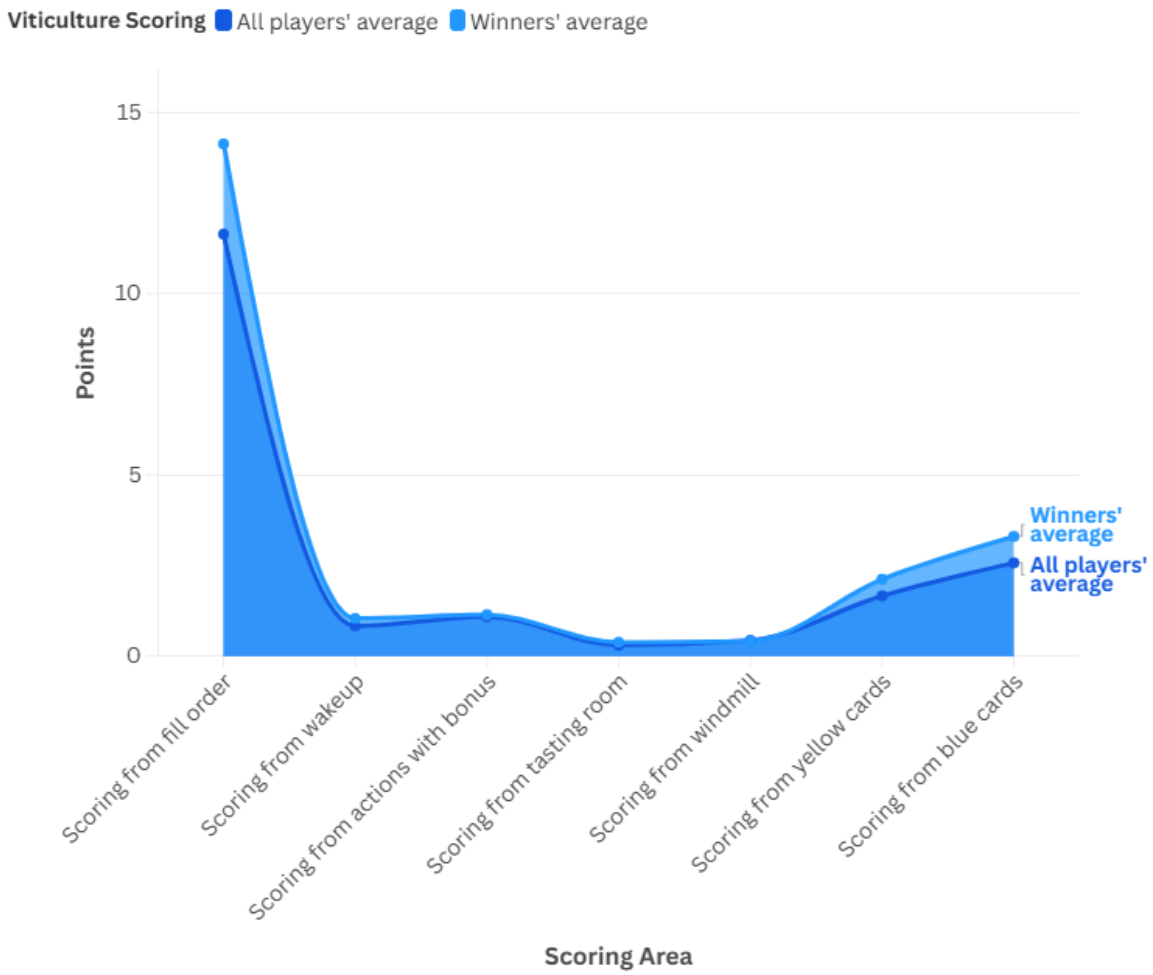
- Line graphs were used to illustrate data separately and combined to show data overtime using *Flourish*.



Key Findings

- Averages were highest on chart 2.
- Averages were lowest on chart 4.

- Line graph with area filled was used to show the contrast of the winners' average and players' average. *Flourish* was using for this visualization.

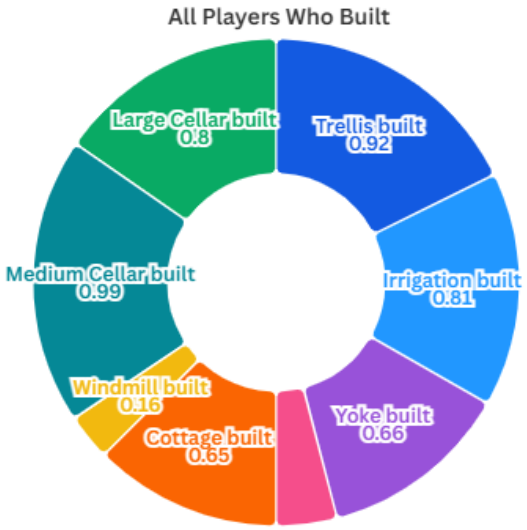


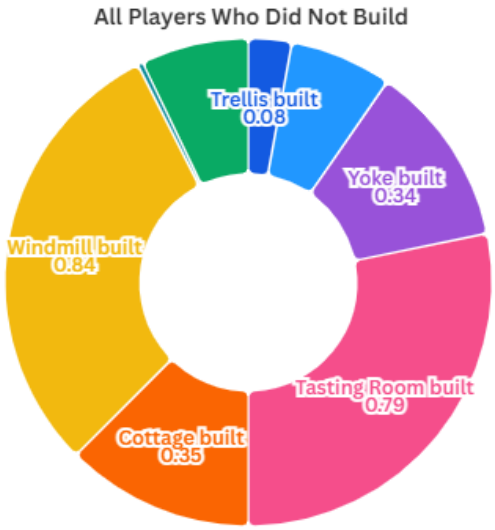
Key Findings

- Fill order yielded the most point which also made the largest impact for winners.
- Tasting room points yielded the least points.

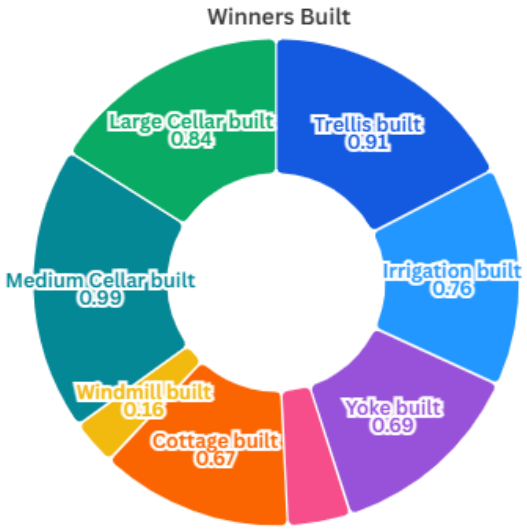
- The Doughnut illustration was used to illustrate the different proportions in a round way and also used color to distinguish different building areas. *Flourish* was used for this visualization.

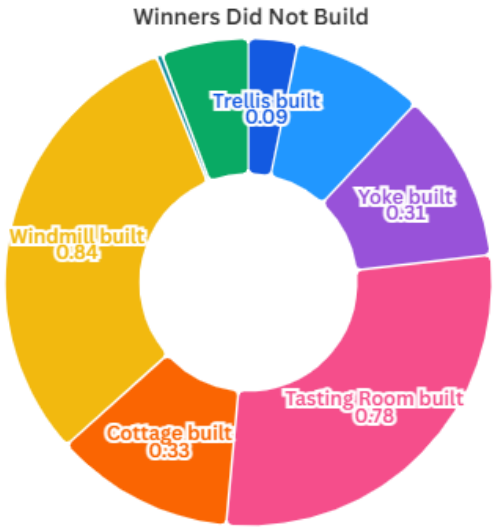
Building





Building

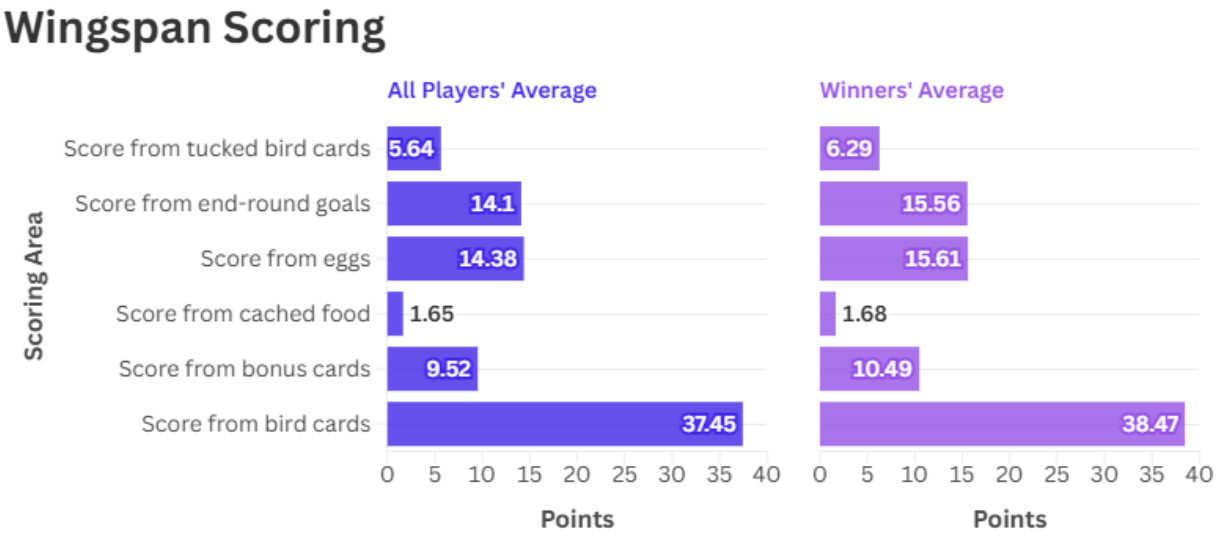




Key Findings

- Most players built a medium cellar.
- Most players did not build a windmill.
- Winners tended to focus slightly more on building yoke while all players average tended to have more irrigation being built.
- Winners also had a higher percentage for large cellar and trellis being built.

- The side by side comparison bar chart was chosen to visually illustrate differences between All Players' average vs. Winners' average. *Flourish* was used to create this visualization.

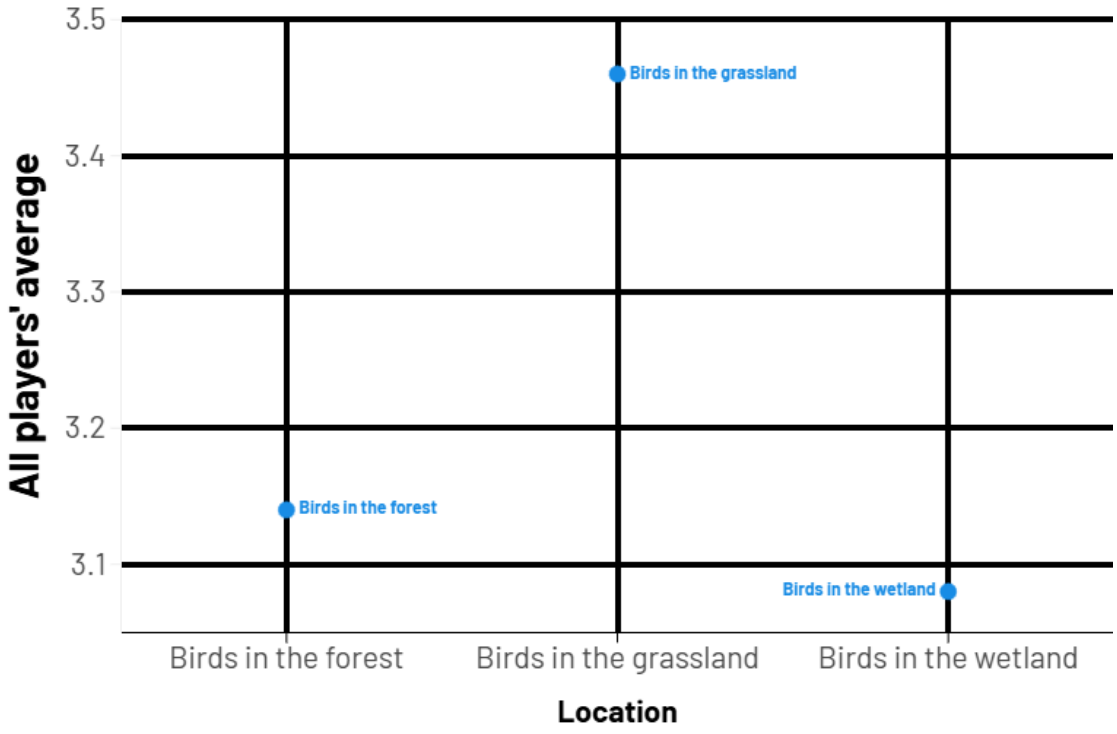


Key Findings

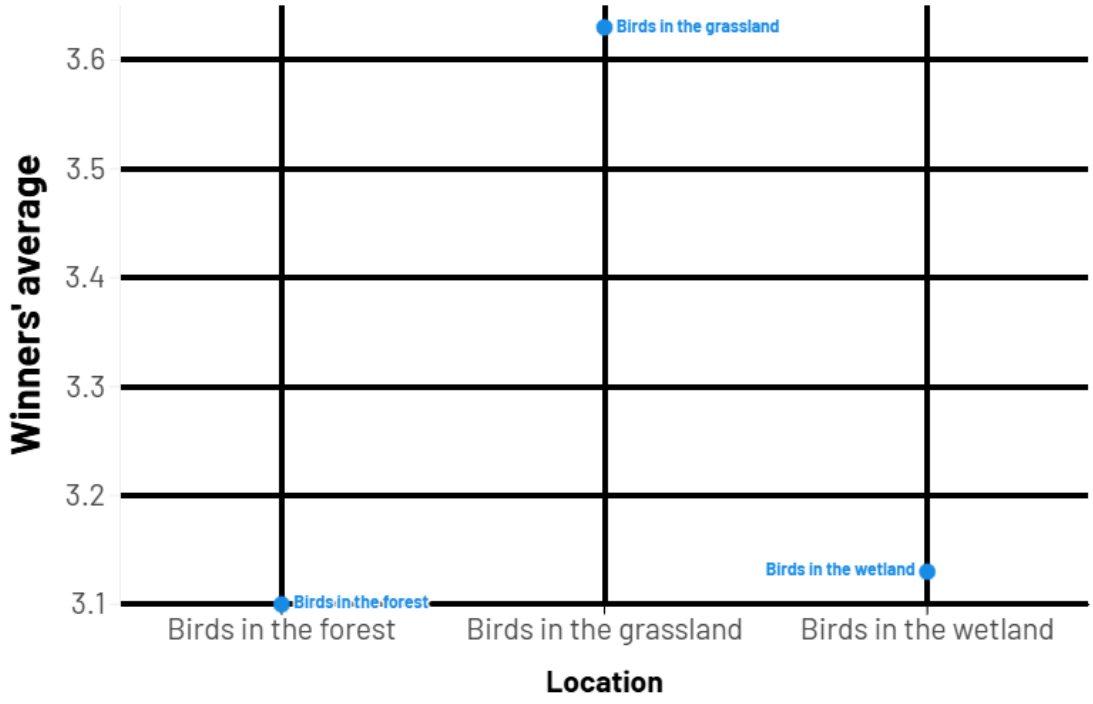
- Both the winners and all players averages received the most points from bird cards
- Winners tended to utilize end-round goals more with receiving 1.46 points more than the all players' average.
- Cached food was the most similar scoring area with only 0.03 difference between the two averages.

- The visualization below is a scatter plot that illustrates the average number of birds per location. *Flourish* was used to create this visualization.

Bird Locations Wingspan

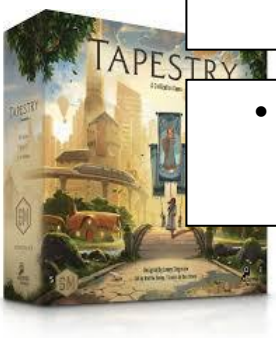


Bird Locations Wingspan



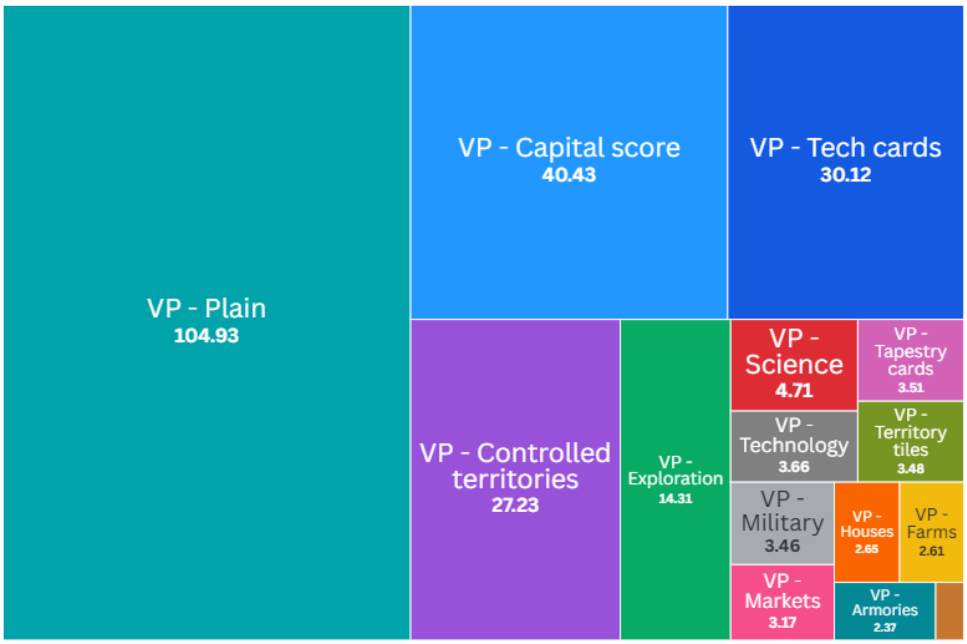
Key Findings

- More birds were placed in the grassland for both all players and winners, but this area was utilized slightly more by winners.
- Winners also had slightly more birds in the wetland than the overall players' average.



- The treemap hierarchy visualization shows a proportional visual for different victory points obtained by all players. The visualization was created using *Flourish*.

All Players' Victory Points

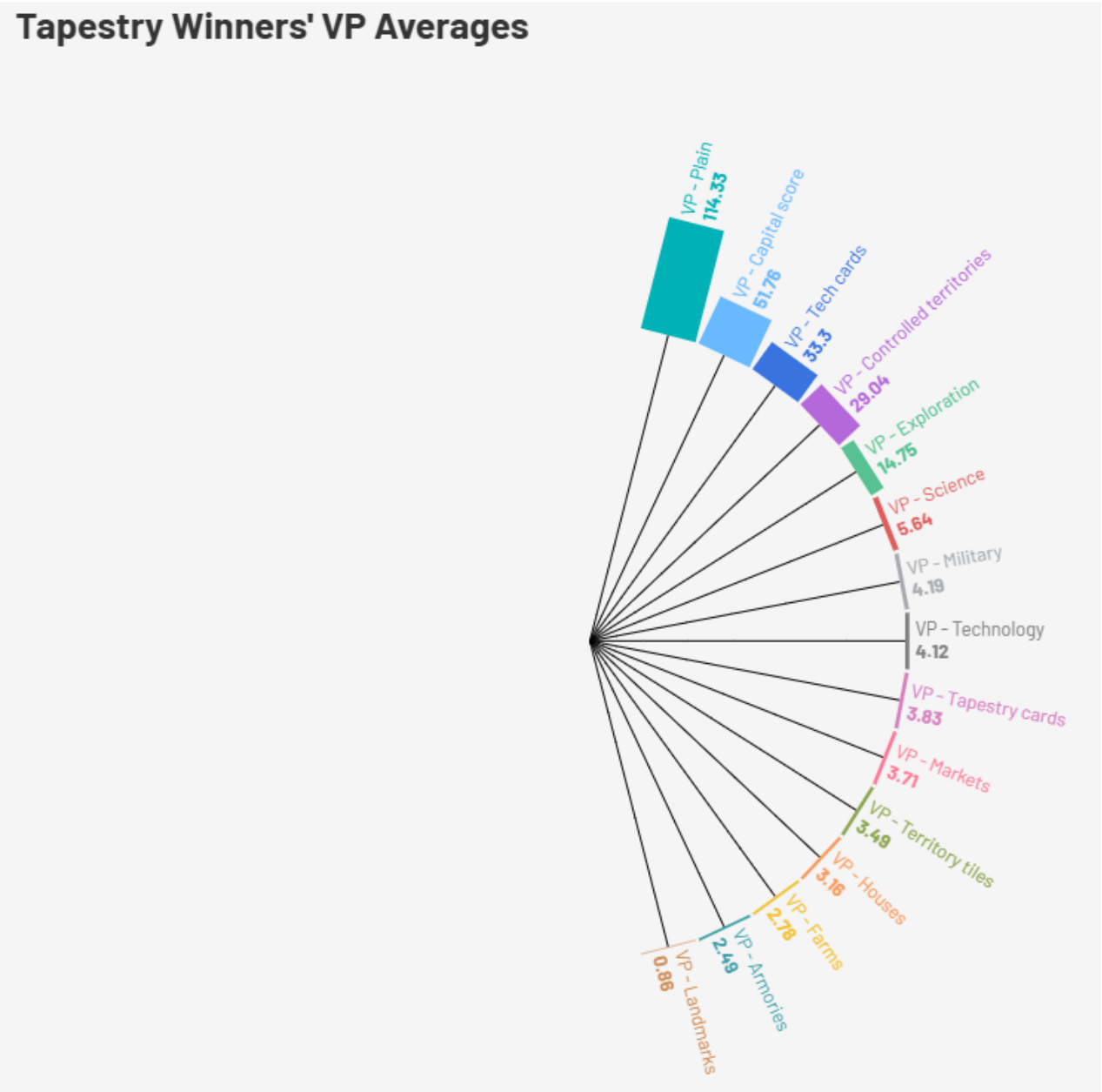


*- Dark Orange- VP- Landmarks- 0.66

Key Findings

- Plain and Capital yielded the most victory points.
- Landmarks yielded the least victory points

- The radial hierarchy visualization shows a proportional visual for different victory points obtained by winners. The visualization was created using *Flourish*.



Key Findings

- Plain and Capital yielded the most victory points.
- Landmarks yielded the least victory points
- Winners received more points in the top three areas (plain, capital, and tech) vs. the all players' averages, respectively.

- An infographic format was created to illustrate key facts. *Canva* and The *Noun Project* were used.

5 Facts about Tapestry

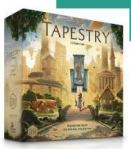
0.97

This is the difference in advances during Era 4 between All Players and the Winners Average.

9.25

Winners' Average Technology Spaces

This was higher than the All players' average by 0.7 who had only 8.55.



0.34

Houses yielded the highest difference between the two groups with All players averaging 4.58 houses and Winners averaging 4.92.

19.98

Difference bwetween Winners' average and All Players Average for income received from income mat.

Winners collected almost 20 mo points from the income mat wit 142.47 compared to 122.49.

277.41

Winners' Total Points

The average points for Winners was 30,24 higher than the average players' average who had 247.17